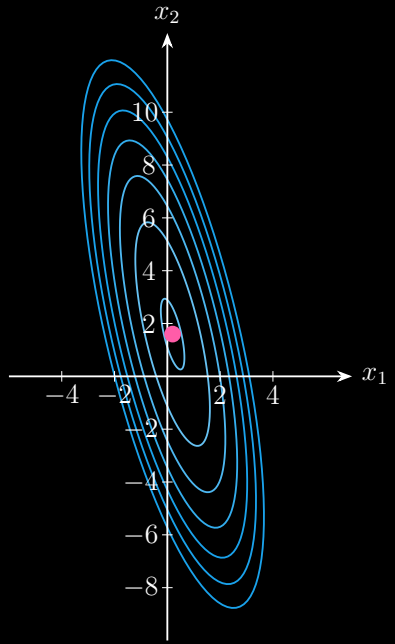
Quadratic optimization:

$$\min_{\boldsymbol{x}} \quad \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{x}$$

with

$$\boldsymbol{A} = \begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$



Quadratic optimization:

$$\min_{\boldsymbol{x}} \quad \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{x}$$

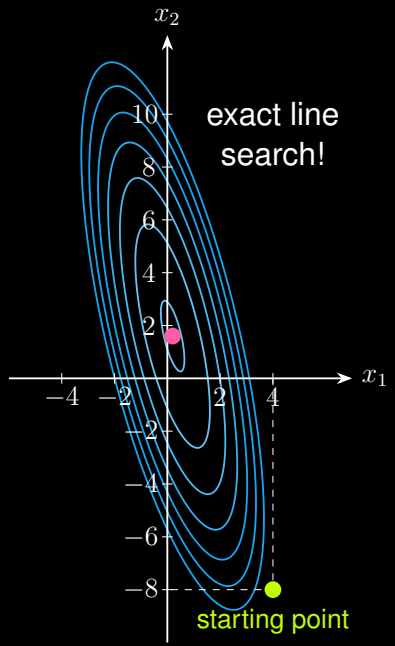
Steepest gradient descent:

$$\boldsymbol{g}_k = \boldsymbol{A}\boldsymbol{x}_k - \boldsymbol{b} \qquad \text{(gradient)}$$

$$\alpha_k = \frac{\boldsymbol{g}_k^\top \boldsymbol{g}_k}{\boldsymbol{g}_k^\top \boldsymbol{A}\boldsymbol{g}_k} \qquad \text{(step size)}$$

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \alpha_k \boldsymbol{g}_k \quad \text{(decision vector)}$$

with an initial point $\boldsymbol{x}_0$.



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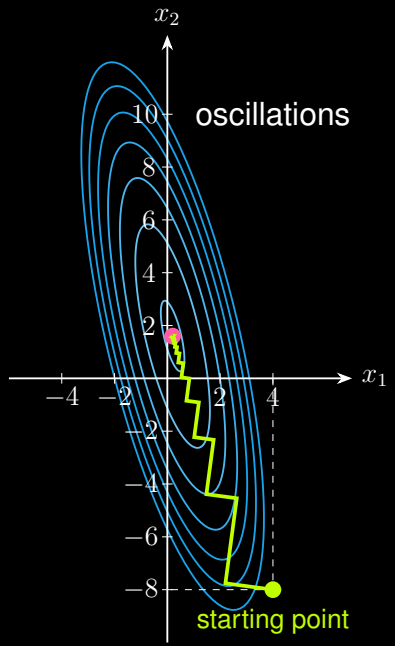

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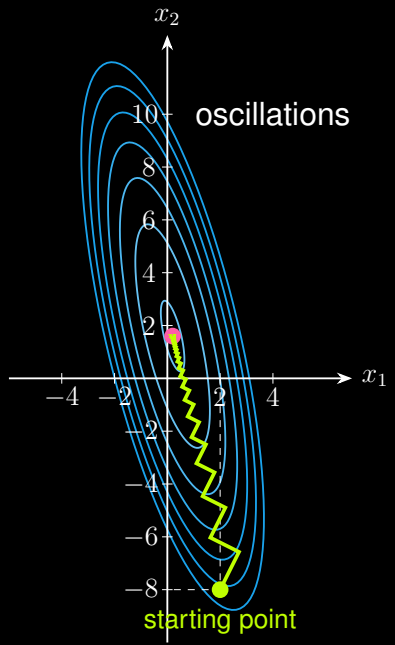

Positive Definite Matrix $\boldsymbol{A} = \begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix}$

Eigenvalue decomposition:

$$\boldsymbol{A} = \lambda_1 \boldsymbol{u}_1 \boldsymbol{u}_1^\top + \lambda_2 \boldsymbol{u}_2 \boldsymbol{u}_2^\top$$

with eigenvalues:

$$\lambda_1 = 9.47 \quad \lambda_2 = 0.53$$

and condition number:

$$\kappa = \frac{\lambda_1}{\lambda_2} = 17.94$$

Positive Definite Matrix $\boldsymbol{A} = \begin{bmatrix} 9 & 2 \\ 2 & 1 \end{bmatrix}$

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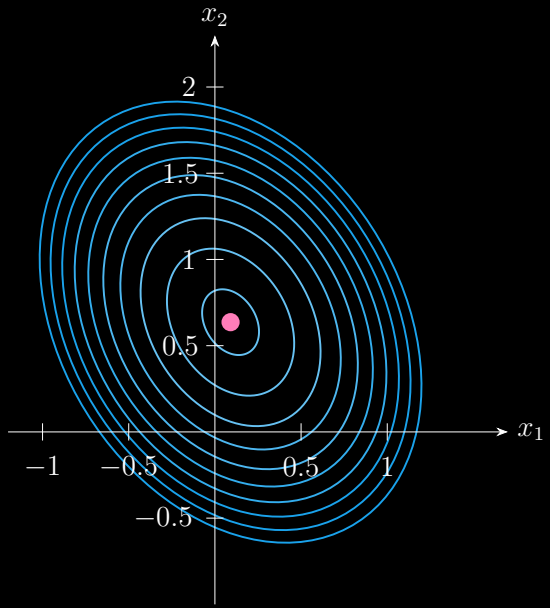
```python
import numpy as np

A = np.array([[9, 2], [2, 1]])
eig_val, eig_vec = np.linalg.eig(A)

print('lambda_1 = ', eig_val[0])
print('lambda_2 = ', eig_val[1])
print('u_1 = ', eig_vec[:, 0])
print('u_2 = ', eig_vec[:, 1])
print('condition number = ',
      eig_val.max() / eig_val.min())
```

Quadratic optimization:

$$\min_{\boldsymbol{x}} \quad \frac{1}{2}\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{x}$$

with

$$\boldsymbol{A} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$



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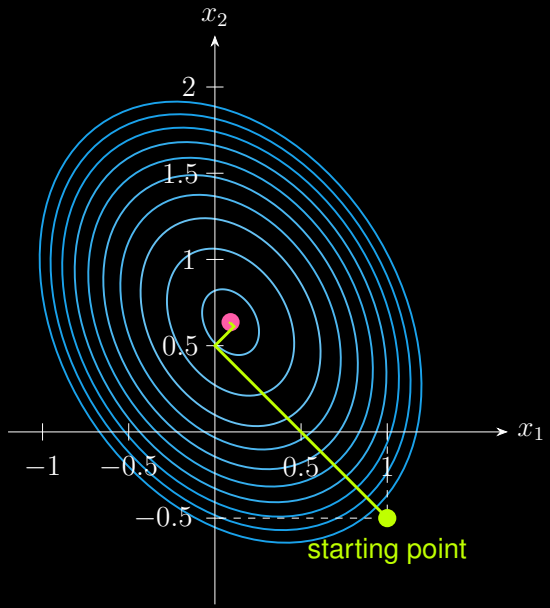

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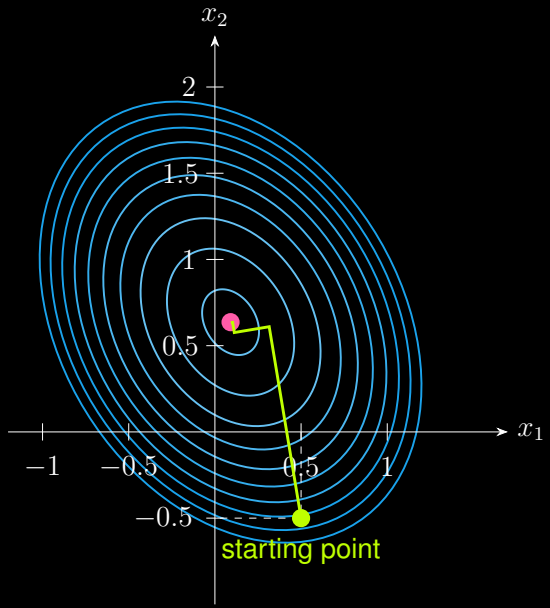


Positive Definite Matrix $\boldsymbol{A} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$

Eigenvalue decomposition:

$$\boldsymbol{A} = \lambda_1 \boldsymbol{u}_1 \boldsymbol{u}_1^\top + \lambda_2 \boldsymbol{u}_2 \boldsymbol{u}_2^\top$$

with eigenvalues:

$$\lambda_1 = 4.62 \qquad \lambda_2 = 2.38$$

and condition number:

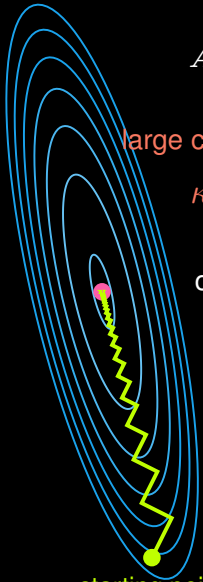
$$\kappa = \frac{\lambda_1}{\lambda_2} = 1.94$$

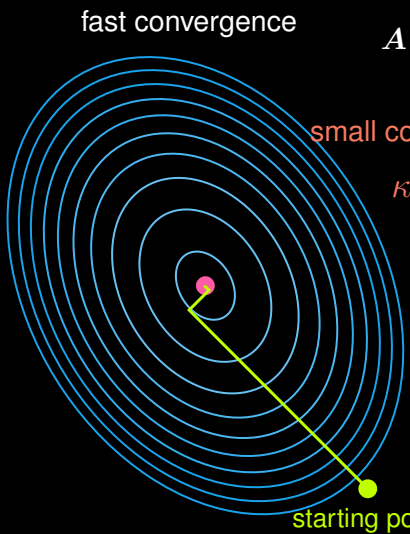
```python
import numpy as np

A = np.array([[4, 1], [1, 3]])
eig_val, eig_vec = np.linalg.eig(A)

print('lambda_1 = ', eig_val[0])
print('lambda_2 = ', eig_val[1])
print('u_1 = ', eig_vec[:, 0])
print('u_2 = ', eig_vec[:, 1])
print('condition number = ',
      eig_val.max() / eig_val.min())
```

Steepest Gradient Descent





Thanks for your attention!

About me:

- Homepage: https://xinychen.github.io
- How to reach me: chenxy346@gmail.com